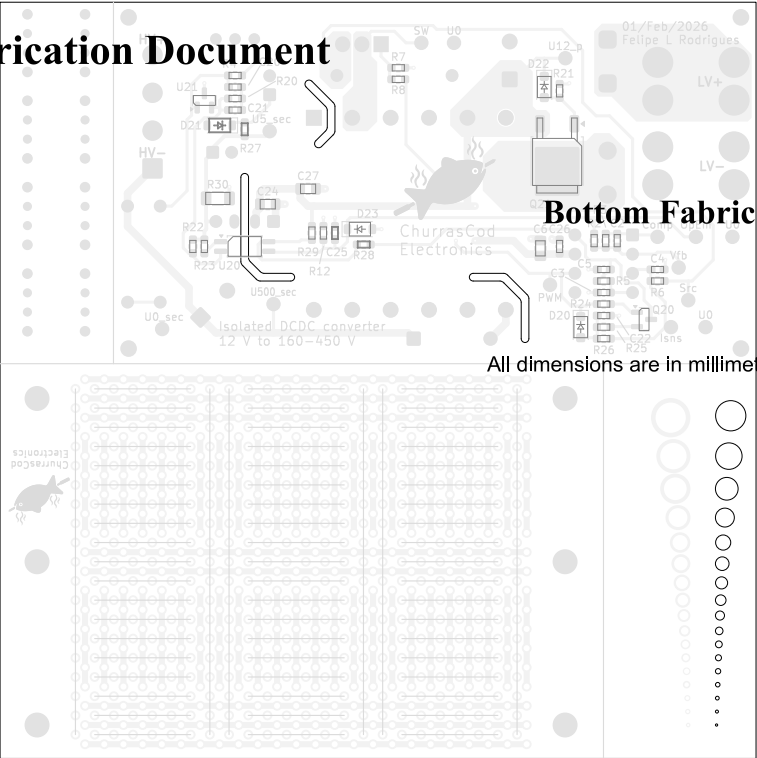


Module Fabrication Document



Bottom Fabrication (Scale 1:1)

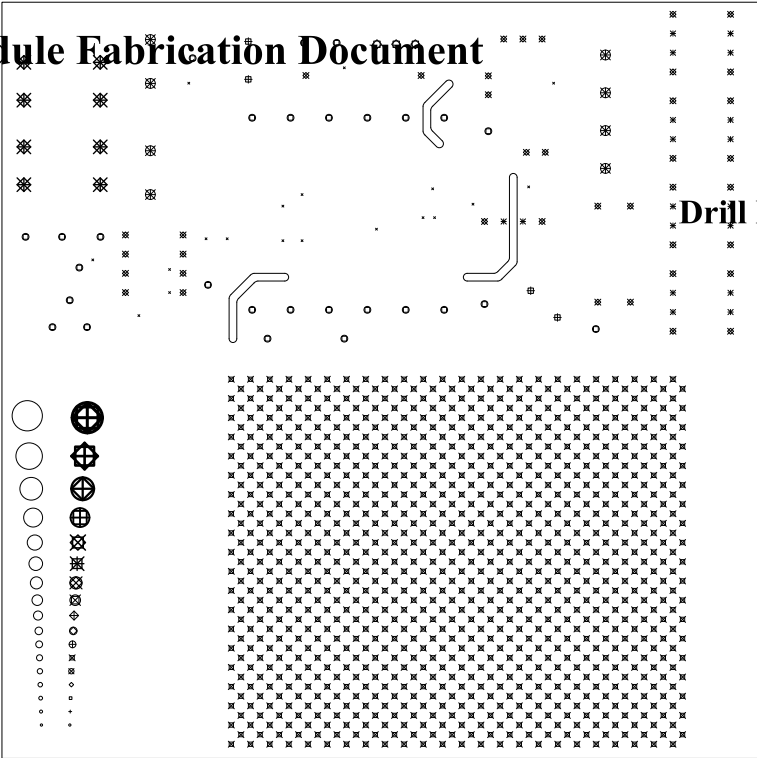
All dimensions are in millimeters unless otherwise specified.

	Comments: Comment 1 Comment 2 Comment 3 Comment 4		Company: FR		Variant: PRELIMINARY	Git Hash: 3c81618
			Board Name: Module		Project Name: Isolated step-up 12:450 V	
	Sheet Title: Bottom Fabrication (Scale 1:1)		File Name: Isolated_12to450V.kicad_pcb	Designer: FR	Date: 2026-02-02	Revision: 1.0.0+ (Unreleased)
	Sheet Path:			Reviewer: NA	Size: A3	Sheet: 2 of 10

Drill Table

Symbol	Count	Hole Size	Plated	Hole Shape	Drill Layer Pair	Hole Type
×	19	0,25mm (9,84mils)	PTH	Round	L1 (Ref) - L2 (Slg, PWR)	Via
○	1	0,30mm (11,81mils)	PTH	Round	L1 (Ref) - L2 (Slg, PWR)	Via
+	1	0,40mm (15,75mils)	PTH	Round	L1 (Ref) - L2 (Slg, PWR)	Via
□	1	0,50mm (19,69mils)	PTH	Round	L1 (Ref) - L2 (Slg, PWR)	Via
◇	1	0,60mm (23,62mils)	PTH	Round	L1 (Ref) - L2 (Slg, PWR)	Via
⊗	1	0,70mm (27,56mils)	PTH	Round	L1 (Ref) - L2 (Slg, PWR)	Via
✱	18	0,70mm (27,56mils)	PTH	Round	L1 (Ref) - L2 (Slg, PWR)	Pad
⊗	937	0,80mm (31,50mils)	PTH	Round	L1 (Ref) - L2 (Slg, PWR)	Via
✱	39	0,80mm (31,50mils)	PTH	Round	L1 (Ref) - L2 (Slg, PWR)	Pad
⊕	1	0,90mm (35,43mils)	PTH	Round	L1 (Ref) - L2 (Slg, PWR)	Via
○	28	0,90mm (35,43mils)	PTH	Round	L1 (Ref) - L2 (Slg, PWR)	Pad
○	1	1,00mm (39,37mils)	PTH	Round	L1 (Ref) - L2 (Slg, PWR)	Via
⊕	4	1,00mm (39,37mils)	PTH	Round	L1 (Ref) - L2 (Slg, PWR)	Pad
⊕	1	1,20mm (47,24mils)	PTH	Round	L1 (Ref) - L2 (Slg, PWR)	Via
○	3	1,20mm (47,24mils)	PTH	Round	L1 (Ref) - L2 (Slg, PWR)	Pad
⊗	8	1,30mm (51,18mils)	PTH	Round	L1 (Ref) - L2 (Slg, PWR)	Pad
⊗	1	1,40mm (55,12mils)	PTH	Round	L1 (Ref) - L2 (Slg, PWR)	Via
⊗	1	1,60mm (62,99mils)	PTH	Round	L1 (Ref) - L2 (Slg, PWR)	Via
✱	1	1,80mm (70,87mils)	PTH	Round	L1 (Ref) - L2 (Slg, PWR)	Via
✱	8	1,80mm (70,87mils)	PTH	Round	L1 (Ref) - L2 (Slg, PWR)	Pad
✱	1	2,00mm (78,74mils)	PTH	Round	L1 (Ref) - L2 (Slg, PWR)	Via
⊕	1	2,50mm (98,43mils)	PTH	Round	L1 (Ref) - L2 (Slg, PWR)	Via
⊕	1	3,00mm (118,11mils)	PTH	Round	L1 (Ref) - L2 (Slg, PWR)	Via
⊕	1	3,50mm (137,80mils)	PTH	Round	L1 (Ref) - L2 (Slg, PWR)	Via
⊕	1	4,00mm (157,48mils)	PTH	Round	L1 (Ref) - L2 (Slg, PWR)	Via
Total 1080						

Module Fabrication Document



Drill Drawing L1 - L4 (Scale 1:1)

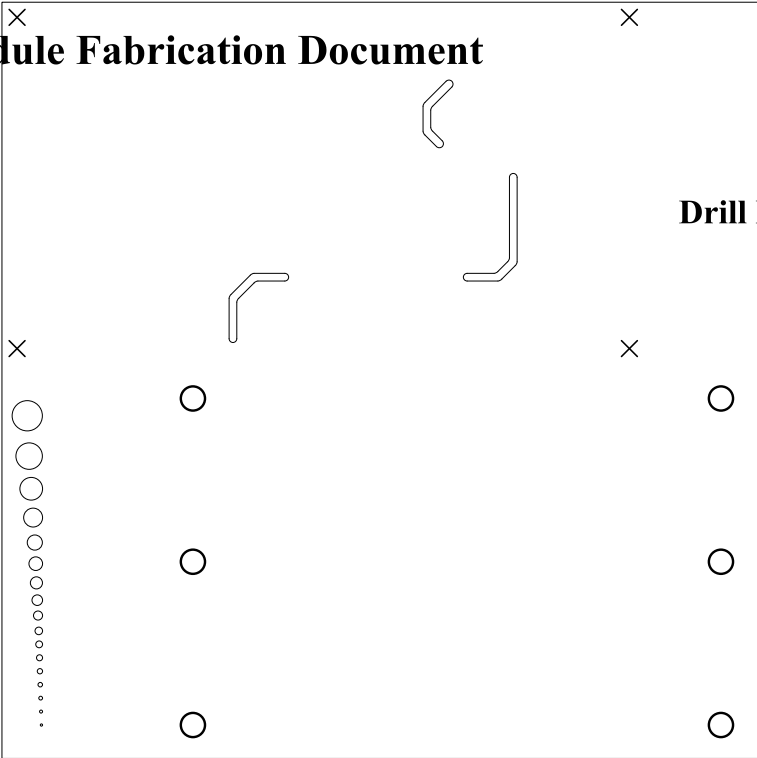
	Comments: Comment 1 Comment 2 Comment 3 Comment 4		Company: FR		Variant: PRELIMINARY	Git Hash: 3c81618
			Board Name: Module		Project Name: Isolated step-up 12:450 V	
	Sheet Title: Drill Drawing (L1 - L4)		File Name: Isolated_12to450V.kicad_pcb	Designer: FR	Date: 2026-02-02	Revision: 1.0.0+ (Unreleased)
	Sheet Path:			Reviewer: NA	Size: A3	Sheet: 3 of 10

Drill Table

Symbol	Count	Hole Size	Plated	Hole Shape	Drill Layer Pair	Hole Type
X	4	2,10mm (82,68mils)	NPTH	Round	L1 (Ref) - L2 (Sig, PWR)	Mechanical
O	6	3,20mm (125,98mils)	NPTH	Round	L1 (Ref) - L2 (Sig, PWR)	Mechanical
	Total 10					

Module Fabrication Document

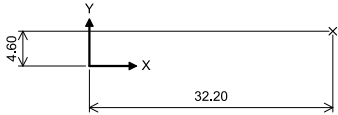
Drill Drawing L1 - L4 (Scale 1:1)



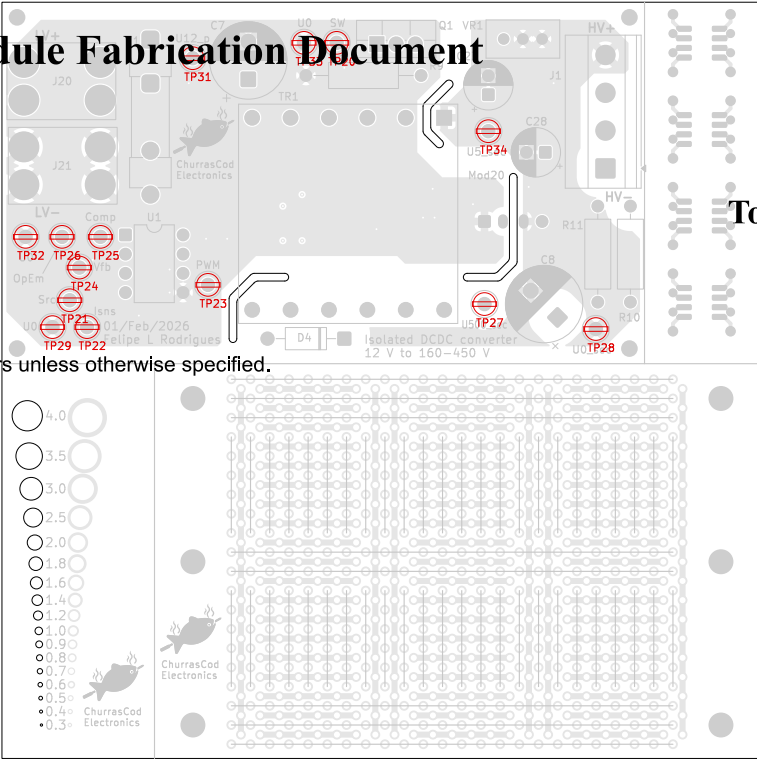
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			Board Name: Module		Project Name: Isolated step-up 12:450 V	
	Sheet Title: Drill Drawing (L1 - L4)		File Name: Isolated_12to450V.kicad_pcb	Designer: FR	Date: 2026-02-02	Revision: 1.0.0+ (Unreleased)
	Sheet Path:			Reviewer: NA	Size: A3	Sheet: 4 of 10

Ref.	Net	X [mm]	Y [mm]
TP20	SW	112.07	-0.77
TP21	Src	76.77	-34.81
TP22	Isns	79.05	-38.36
TP23	PWM	95.05	-32.78
TP24	Vfb	78.04	-30.49
TP25	Comp	80.83	-26.43
TP26	OpEm	75.75	-26.43
TP27	U500_sec	131.63	-35.32
TP28	U0_sec	146.36	-38.62
TP29	U0	74.48	-38.36
TP31	U12_p	93.02	-2.80
TP32	U0	70.92	-26.43
TP33	U0	107.75	-0.77
TP34	U5_sec	132.14	-12.46

Ref.	Net	X [mm]	Y [mm]
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Module Fabrication Document



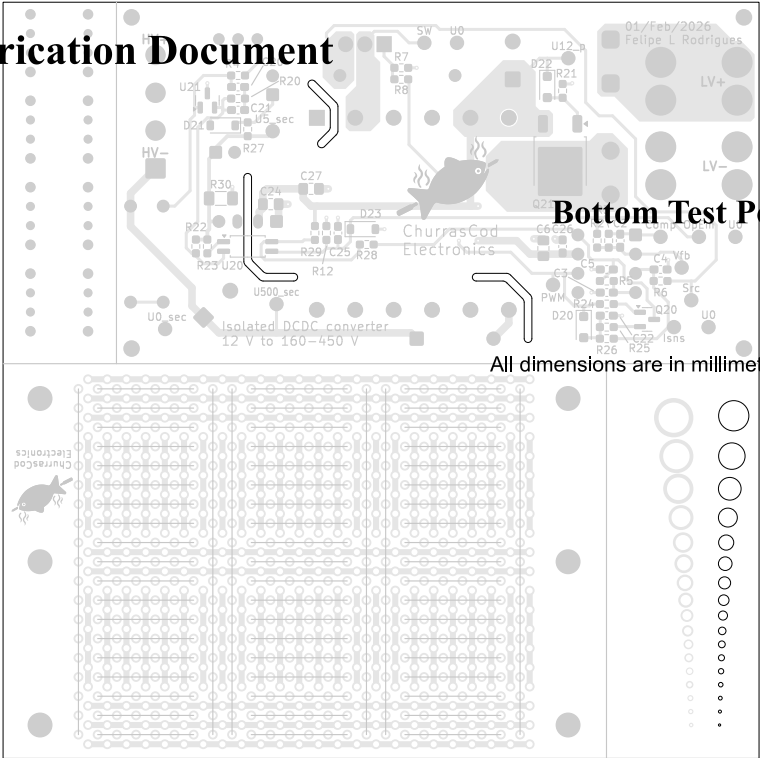
Top Test Points (Scale 1:1)

All dimensions are in millimeters unless otherwise specified.

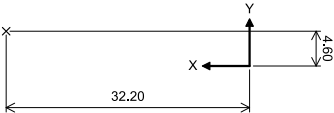
	Comments: Comment 1 Comment 2 Comment 3 Comment 4		Company: FR		Variant: PRELIMINARY	Git Hash: 3c81618
			Board Name: Module		Project Name: Isolated step-up 12:450 V	
	Sheet Title: Top Test Points (Scale 1:1)		File Name: Isolated_12to450V.kicad_pcb	Designer: FR	Date: 2026-02-02	Revision: 1.0.0+ (Unreleased)
	Sheet Path:			Reviewer: NA	Size: A3	Sheet: 5 of 10

Ref.	Net	X [mm]	Y [mm]
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Module Fabrication Document



Bottom Test Points (Scale 1:1)

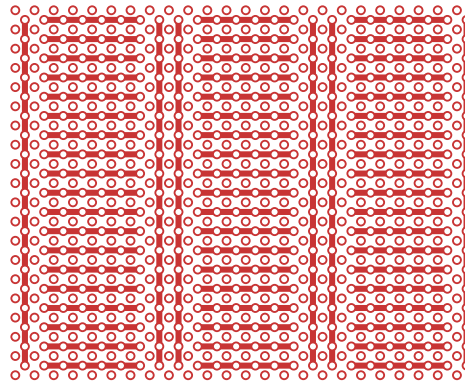


All dimensions are in millimeters unless otherwise specified.

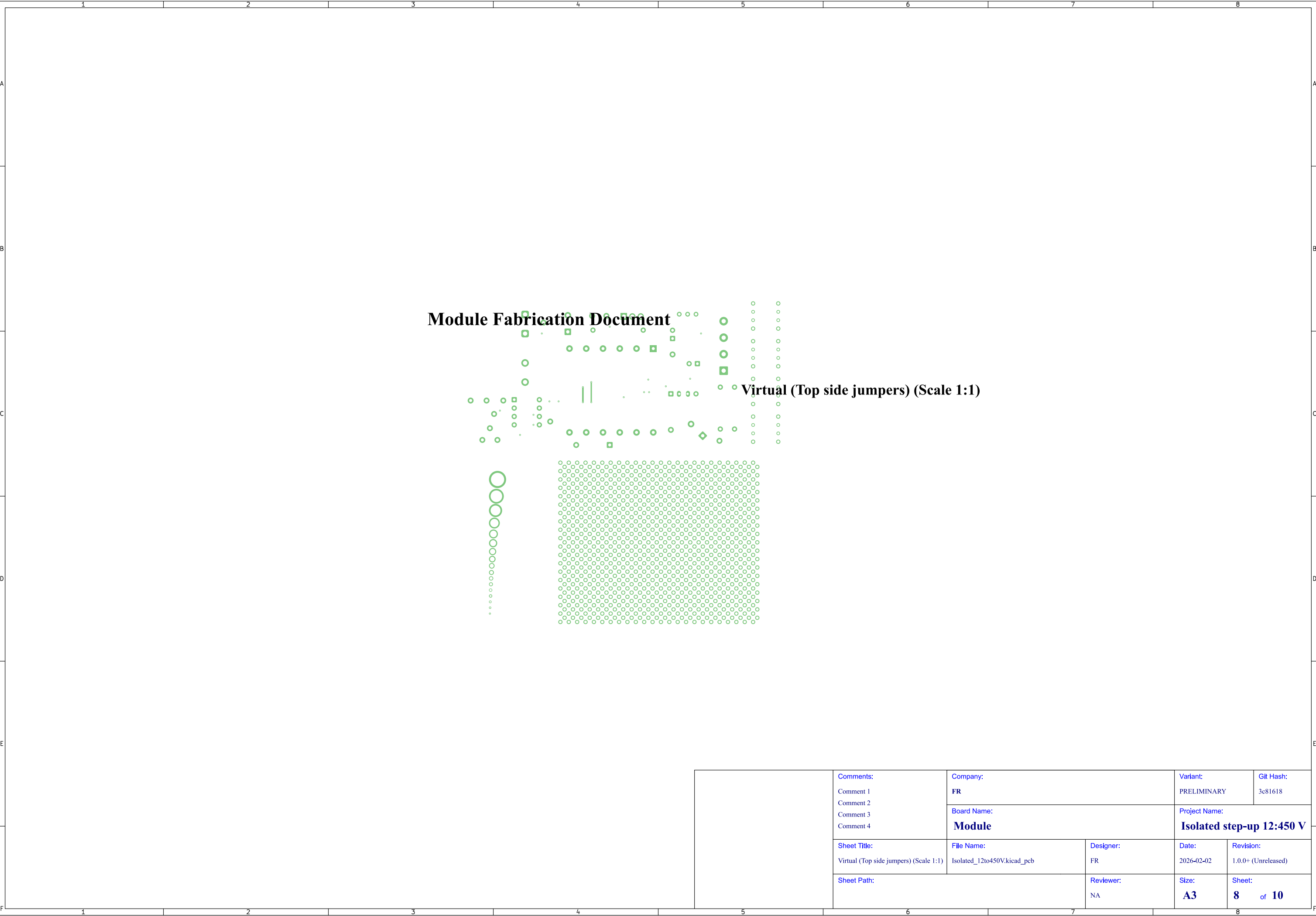
	Comments: Comment 1 Comment 2 Comment 3 Comment 4		Company: FR		Variant: PRELIMINARY	Git Hash: 3c81618
			Board Name: Module		Project Name: Isolated step-up 12:450 V	
	Sheet Title: Bottom Test Points (Scale 1:1)		File Name: Isolated_12to450V.kicad_pcb	Designer: FR	Date: 2026-02-02	Revision: 1.0.0+ (Unreleased)
	Sheet Path:			Reviewer: NA	Size: A3	Sheet: 6 of 10

Rule Fabrication Document

A decorative graphic consisting of two vertical columns of red, circuit-like lines. Each line starts with a small circle at the top and branches out into three horizontal segments before ending in another small circle at the bottom. The lines are arranged in a staggered, repeating pattern.

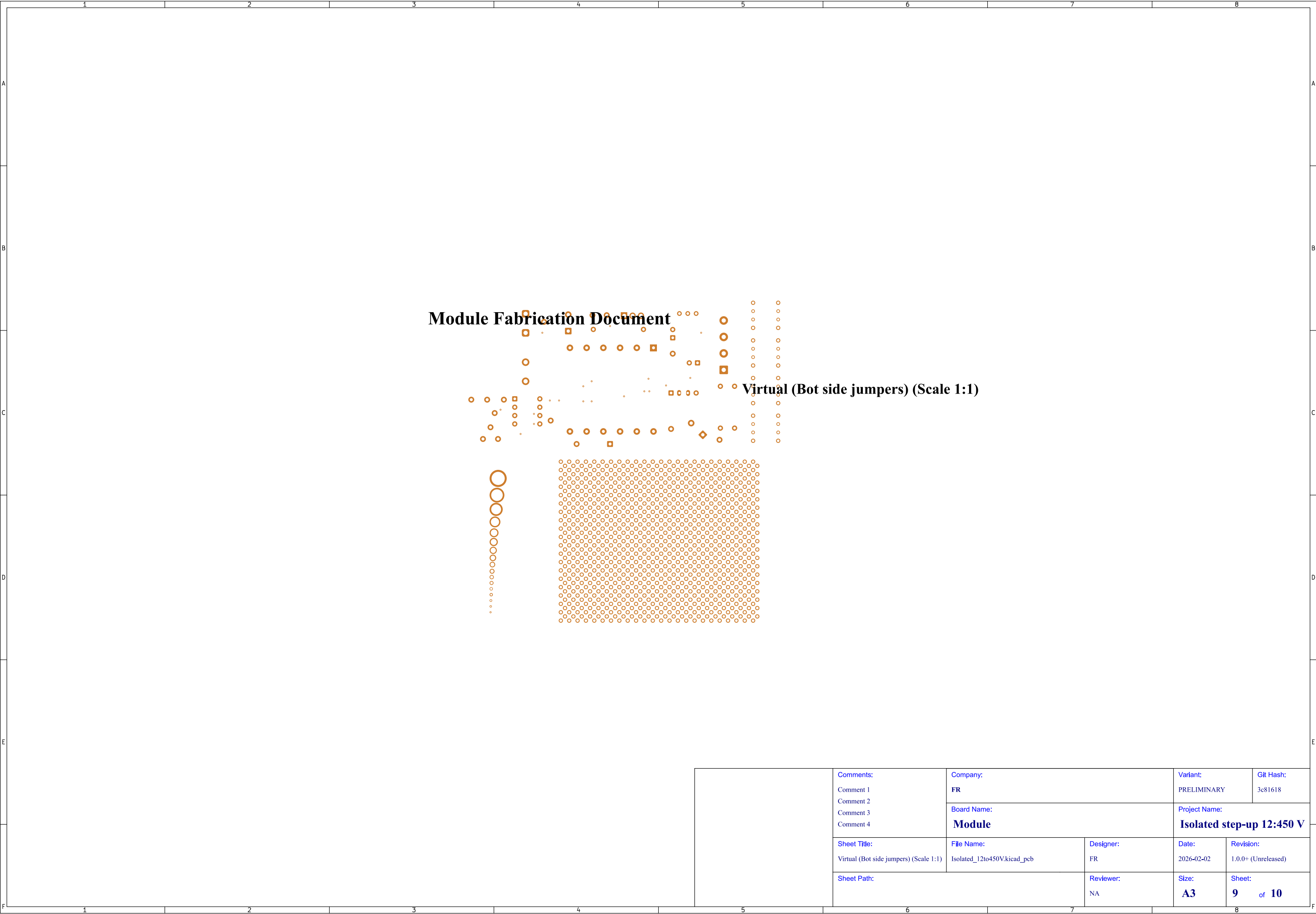


Comments: Comment 1 Comment 2 Comment 3 Comment 4	Company: FR		Variant: PRELIMINARY	Git Hash: 3c81618	
	Board Name: Module		Project Name: Isolated step-up 12:450 V		
	Sheet Title: L1 (Ref) (Scale 1:1)	File Name: Isolated_12to450V.kicad_pcb	Designer: FR	Date: 2026-02-02	Revision: 1.0.0+ (Unreleased)
	Sheet Path:		Reviewer: NA	Size: A3	Sheet: 7 of 10



Module Fabrication Document

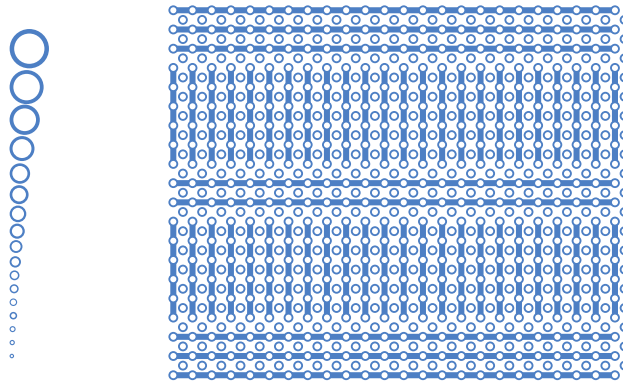
Virtual (Top side jumpers) (Scale 1:1)



Module Fabrication Document

Virtual (Bot side jumpers) (Scale 1:1)

Module Fabrication Document



	Comments: Comment 1 Comment 2 Comment 3 Comment 4	Company: FR	Variant: PRELIMINARY		Git Hash: 3c81618
		Board Name: Module	Project Name: Isolated step-up 12:450 V		
	Sheet Title: L2 (Sig, PWR) (Scale 1:1)	File Name: Isolated_12to450V.kicad_pcb	Designer: FR	Date: 2026-02-02	Revision: 1.0.0+ (Unreleased)
	Sheet Path:		Reviewer: NA	Size: A3	Sheet: 10 of 10